Mason Holcombe HW2 Stat 488

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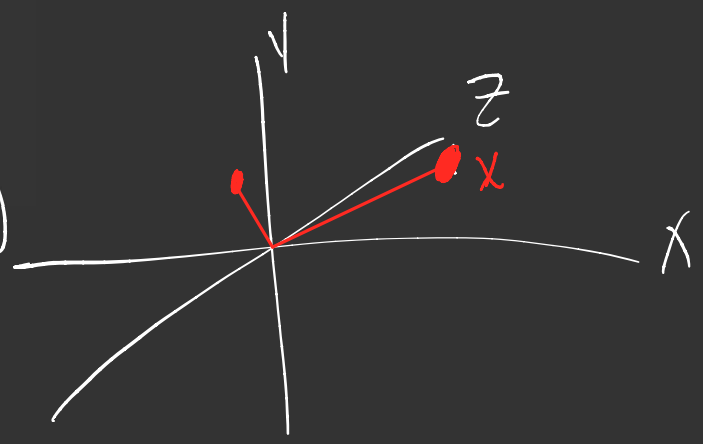
a)



$x' = [5, 1, 3]$

$y' = [-1, 3, 1]$

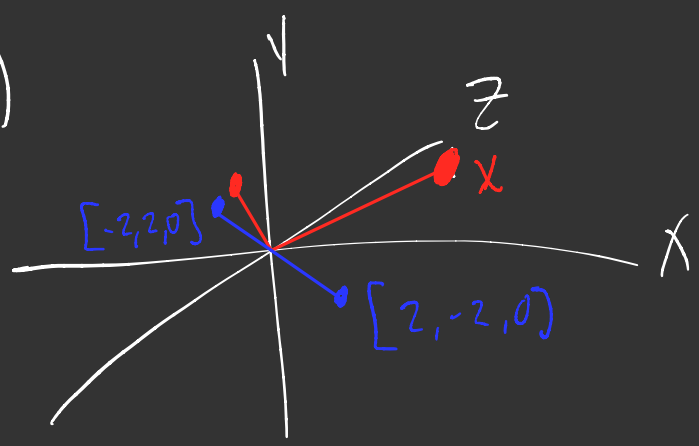
b) $L(x) = \sqrt{5^2 + 1^2 + 3^2} = \sqrt{35}$

$$\theta = \cos^{-1}\left(\frac{x^T y}{L_x L_y}\right) = \cos^{-1}\left(\frac{1}{\sqrt{35} \cdot \sqrt{11}}\right)$$

$$= 87.08^\circ$$

$$\text{Proj}_{yx} = \frac{y^T x}{x^T x} x = \frac{1}{35}[-1, 3, 1]$$

c)



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$$A = \begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$$

a) yes, A is symmetric

$$A = A^T = \begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$$

b) $\det\left(\begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) = 0$

$$(9-\lambda)(6-\lambda) - 4 = 0$$

$$\lambda^2 - 15\lambda + 50 = 0$$

$$(\lambda - 10)(\lambda - 5) = 0$$

$$\lambda = 10, 5$$

Both > 0, so A is positive definite

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$$A = \begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$$

a) $\lambda_1 = 10$, $\lambda_2 = 5$

$$v_1 = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

v_1

$(A - 10I)x = 0$

$$\begin{pmatrix} -1 & -2 \\ -2 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$-x - 2y = 0$
$-2x - 4y = 0$ $\Rightarrow$ $x = -2y$

v_2

$(A - 5I)x = 0$

$$\begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$\Rightarrow$ $4x - 2y = 0$
$-2x + y = 0$ $\Rightarrow$ $4x = 2y$, $2x = y$

b)

$$A = 10\begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}\begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} + 5\begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}\begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$= 10\begin{bmatrix} 4/5 & -2/5 \\ -2/5 & 1/5 \end{bmatrix} + 5\begin{bmatrix} 1/5 & 2/5 \\ 2/5 & 4/5 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -4 \\ -4 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix} \checkmark$$

c) $$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \frac{1}{50}\begin{pmatrix} 6 & 2 \\ 2 & 9 \end{pmatrix}$$

d) $\det(A - \lambda I) = 0$

$$= \left| \begin{pmatrix} 6/50 & 2/50 \\ 2/50 & 4/50 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

$$= \left(6/50 - \lambda\right)\left(4/50 - \lambda\right) - \left(\frac{2}{50}\right)^2$$

$$= \lambda^2 - \frac{3}{10}\lambda + 1/50 = 0$$

$$\Rightarrow 50\lambda^2 - 15\lambda + 1 = 0$$

$$\lambda_1 = 1/5, \quad \lambda_2 = 1/10$$

$$v_1 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$\underline{v_1}$

$$(A - 1/5\, I)x = 0$$

$$= \frac{1}{50}\begin{pmatrix} -4 & 2 \\ 2 & -1 \end{pmatrix}$$

$$\Rightarrow -4x + 2y = 0$$

$$\Rightarrow y = 2x$$

$\underline{v_2}$

$$\left(A - \frac{1}{10} I\right)x = 0$$

$$= \frac{1}{50}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$\Rightarrow x + 2y = 0$$

$$\Rightarrow x = -2y$$

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$$Q = 3x_1^2 + 3x_2^2 - 2x_1x_2$$

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 3-\lambda & -1 \\ -1 & 3-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)^2 - 1 = 0$$

$$\lambda^2 - 6\lambda + 8 = 0$$

$$(\lambda - 4)(\lambda - 2) = 0$$

$$\lambda_1 = 2, \lambda_2 = 4$$

both positive, so Q is positive definite

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$$\Sigma_x = \begin{bmatrix} 3 & & & 0 \\ & 3 & & \\ & & 3 & \\ 0 & & & 3 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}$$

$$\mu_x^T = [3, 2, -2, 0]$$

a) $$E(Ax) = \underset{3\times 4}{A}\ \underset{4\times 1 = 3\times 1}{\mu_x}$$

$$= \begin{bmatrix} 1 & 9 & 3 \end{bmatrix}$$

b) $$\mathrm{cov}(Ax) = \underset{3\times 4}{A}\ \underset{4\times 4}{\Sigma_x}\ \underset{4\times 3 = 3\times 3}{A^T}$$

$$= \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

c) All three pairs of linear combinations from $y = Ax$ are have zero covariance

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$$\Sigma_x = \begin{bmatrix} 25 & -2 & 4 \\ -2 & 4 & 1 \\ 4 & 1 & 9 \end{bmatrix}$$

a) $V^{1/2} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $(V^{1/2})^{-1} = \begin{bmatrix} 1/5 & & 0 \\ & 1/2 & \\ 0 & & 1/3 \end{bmatrix}$

$$\rho = (V^{1/2})^{-1} \Sigma_x (V^{1/2})^{-1}$$

$$= \begin{bmatrix} 1 & -1/5 & 4/15 \\ -1/5 & 1 & 1/6 \\ 4/15 & 1/6 & 1 \end{bmatrix}$$

b) $V^{1/2} \rho V^{1/2} =$

$$= \begin{bmatrix} 25 & -2 & 4 \\ -2 & 4 & 1 \\ 4 & 1 & 9 \end{bmatrix} = \Sigma_x \checkmark$$

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$$\Sigma_X = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix} \qquad A = \begin{bmatrix} 1, & -1 \end{bmatrix}$$

$$\mu_X^T = [4,3,2,1] \qquad B = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$$

Partition X as $\left[\begin{array}{c} x_1 \\ x_2 \\ \hline x_3 \\ x_4 \end{array}\right] = \left[\begin{array}{c} x^{(1)} \\ \hline x^{(2)} \end{array}\right]$

a) $\mathbb{E}(x^{(1)}) = [4,3]$

b) $\mathbb{E}(Ax^{(1)}) = [1 \ -1]\begin{bmatrix} 4 \\ 3 \end{bmatrix} = 1$

c) $\operatorname{cov}(x^{(1)}) = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

d) $\operatorname{cov}(Ax^{(1)}) = A\Sigma_{x^{(1)}}A^T$

$= [1 \ -1]\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$= 4$

e) $\mathbb{E}(x^{(2)}) = [2 \ 1]$

f) $\mathbb{E}(Bx^{(2)}) = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

g) $\operatorname{cov}(x^{(2)}) = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$

h) $\operatorname{cov}(Bx^{(2)}) = B\Sigma_{x^{(2)}}B^T$

$= \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$

$= \begin{bmatrix} 48 & -8 \\ -8 & 4 \end{bmatrix}$

i) $\operatorname{cov}(x^{(1)}, x^{(2)}) = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$

j) $\operatorname{cov}(Ax^{(1)}, Bx^{(2)}) =$

$A\Sigma_{x^{(1,2)}}B^T$

$= [1 \ -1]\begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$

$= [1 \ 2]\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$

$= [0 \ 2]$